A Model of Bank Failures: Funding Frictions and the Dynamics Before Collapse

João Rudge Leite

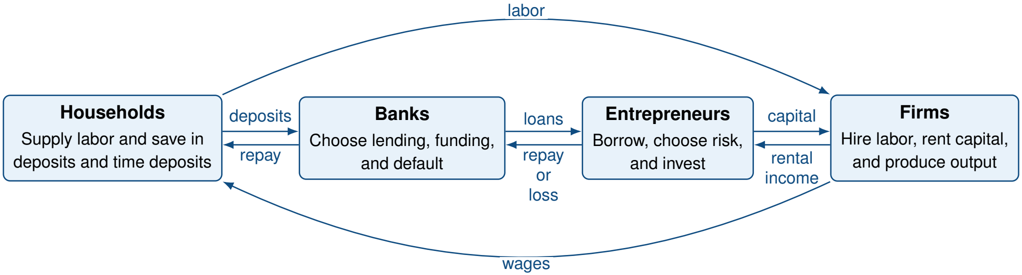
Penn State

DSE 2026 Dynamic Finance and Banking
University of Michigan
August 6, 2026

What this paper does

- Empirical motivation:
  - Before failure, banks increase leverage and rely more on time deposits.
  - Profitability declines while loan losses rise closer to failure.
- Question: How do funding (deposits vs time deposits) and debt dilution shape the buildup of bank distress and eventual failure?
- Mechanism:
  - Time deposits reduce rollover risk and the need for inefficient asset liquidation.
  - Debt dilution encourages additional debt issuance as banks weaken.
- Result: Liability-side incentives help explain the gradual balance-sheet deterioration preceding bank failure.

Model environment



- Banks face liquidity shocks, limited commitment, and self-fulfilling runs.
- Banks cannot issue equity, and reducing lending entails liquidation costs.
- Higher lending rates induce entrepreneurs to choose riskier capital projects.
- Bank default occurs when continuation becomes infeasible.

Bank problem: the relevant tradeoff

$$V^c(z,n,\ell,b) = \max_{R'_\ell, d', b'} \left\{ \underbrace{\text{div}}_{\text{today}} + \underbrace{\mathbb{E}\left[\sigma \Lambda h(z',n',\ell',b',\xi') V^c(z',n',\ell',b')\right]}_{\text{future}} \right\}$$

Bank choices

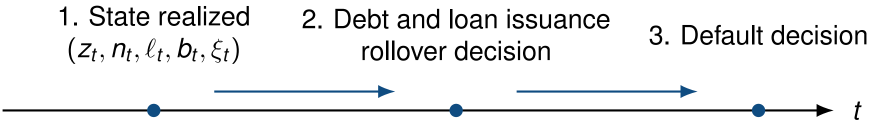
- Lending rate
  set their rate taking $\ell^d(R'_\ell, z, r^e)$
- Deposits
  mature every period
- Time deposits
  pay coupon, fraction λ mat. every period

Key frictions

- Non-negative dividends
- Positive expected equity
- Endogenous debt prices
- Self-fulfilling runs
- Costly liquidation of assets
- Issuance and buyback costs for deposits

Funding choices affect current payouts but also shape future borrowing costs and default risk.

Timing creates rollover risk



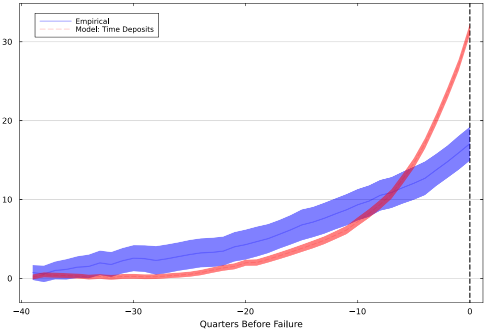
- Creditors price new funding before the default decision is realized.
- Pessimistic rollover beliefs can lower debt prices and validate default.

Model validation

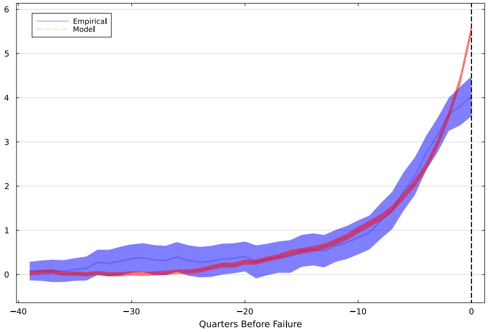
- Parameters are calibrated to match steady-state moments.
- I evaluate whether the model reproduces the untargeted dynamics preceding bank failures.
- The comparison focuses on:
  - funding composition,
  - leverage,
  - profitability and loan losses (appendix).

The transition toward failure is not targeted in the calibration.

Model validation



Time deposits



Leverage

Counterfactuals

No time deposits

- Eliminates long-term funding.
- Increases rollover risk.
- Isolates the role of funding maturity.

No debt dilution

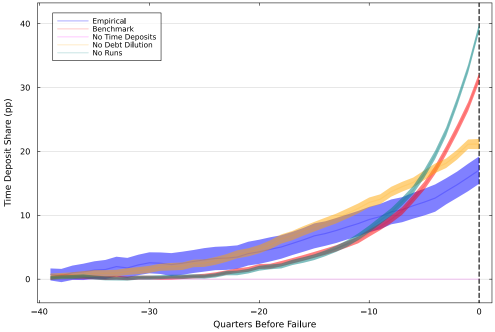
- Covenant protects existing creditors.
- Eliminates debt dilution.
- Isolates leveraging incentives.

No runs

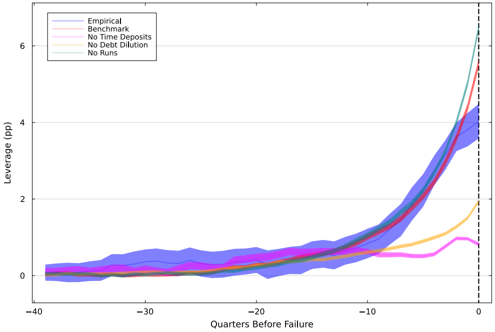
- Eliminates self-fulfilling runs.
- Removes refinancing crises.
- Isolates leverage incentives.

Which mechanism explains the pre-failure dynamics?

Counterfactual: debt dilution



Time deposits



Leverage

- Eliminating debt dilution reduces leverage and, consequently, banks' reliance on time deposits despite their higher market value.

Aggregate implications

- Eliminating debt dilution generates the largest stability gain: failures fall by about 23 percent relative to the benchmark.
- Eliminating self-fulfilling runs has a smaller effect on aggregate outcomes.
- Eliminating time deposits increases failures by roughly fourfold, highlighting their stabilizing role.
- Aggregate capital, output, consumption, and welfare move little across counterfactuals.

Long-term funding stabilizes banks without substantially changing aggregate lending.

Conclusion

- Banks approaching failure gradually shift toward time deposits, increase leverage, and experience worsening profitability and loan performance.
- Long-term funding has a dual role: it reduces rollover risk but creates debt-dilution incentives.
- Debt dilution primarily distorts liability choices; self-fulfilling runs and asset liquidation primarily affect rollover risk.
- Takeaway: liability-side incentives are important to understand the path from distress to failure.

Appendix

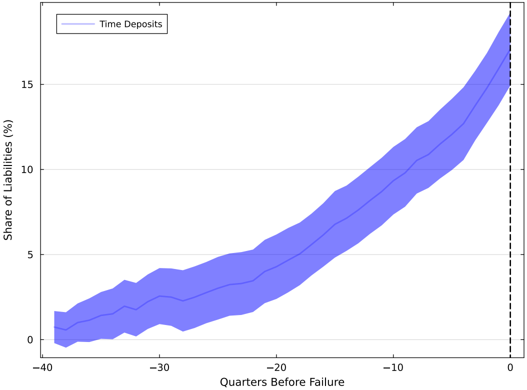
Backup: event-study specification

$$y_{j,t} = \alpha_j + \alpha_t + \sum_{k=-40}^{0} \mathbb{I}_{k=t}\beta_k + \varepsilon_{j,t}.$$

- k indexes quarters to failure.
- The omitted period is $k = -40$, ten years before failure.
- Coefficients measure the evolution of failing banks relative to banks that never failed in the sample.

Back to overview

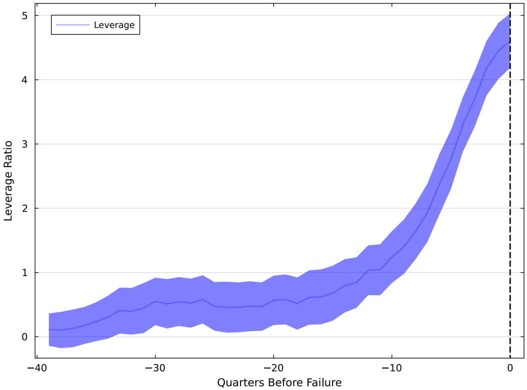
Empirical motivation: banks rely more on long-term funding



- Time deposits increase steadily as banks approach failure.
- Banks substitute toward longer-maturity funding.

Back to overview

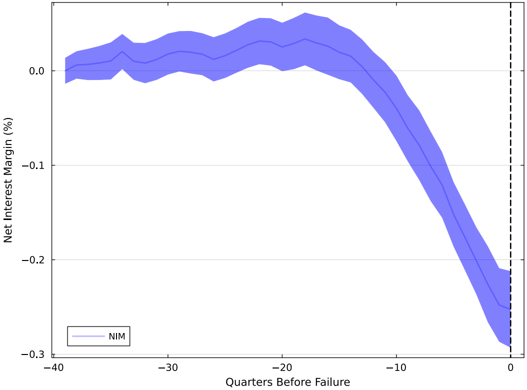
Empirical motivation: banks increase leverage



- Leverage rises persistently before default.

Back to overview

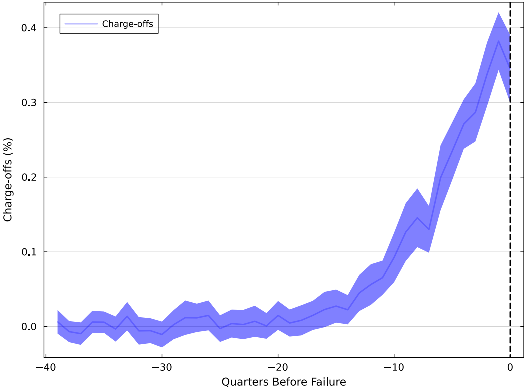
Empirical motivation: profitability deteriorates



- Net interest margins compress before failure.
- Lower profitability slows the rebuilding of bank net worth.

Back to overview

Empirical motivation: credit losses increase



- Loan losses rise sharply as banks approach failure.
- Deteriorating asset quality precedes default by several quarters.

▸ Back to overview

Calibrated Parameters

Table 1: Calibrated Parameters

Parameter	Description	Calibrated Value
σ_z	Std. dev. of credit-quality shock	0.9
σ_ω	Std. dev. of i.i.d. shock	0.0175
ϕ	Fixed operating cost	10^{-3}
ζ	Deposit issuance cost	9×10^{-4}
θ	Shape of loan demand	1.9
σ	Manager myopia	0.98
ψ_0	Success-probability intercept	1.645
ψ_1	Success-probability sensitivity to risk	324
η	Conditional run probability	0.01
γ	Recovery rate on assets	0.9
$r^e_\star$	Steady-state entrepreneurial return	0.017
δ	Depreciation rate	0.0143
$\bar{n}$	Equity of new banks	0.02
$\bar{k}$	Loan amount	98.95
ϑ	Scale of the Loan demand	0.011

▸ Back to validation

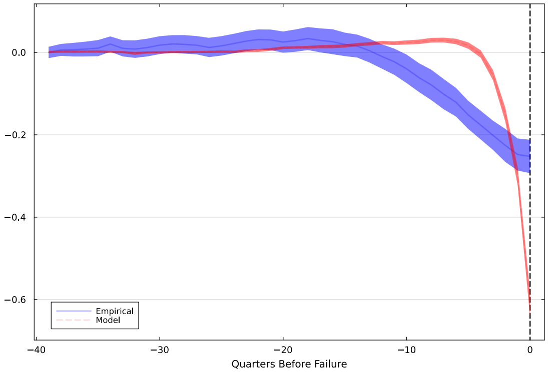
Targeted Moments

Back to validation

Table 2: Targeted Moments: Data vs. Model

Moment	Description	Data (%)	Model (%)
Means			
$\mathbb{E}[r^e x]$	Entrepreneurial return	2.75	2.75
$\mathbb{E}[1-h]$	Share of failing banks	0.10	0.10
$\mathbb{E}\left[\frac{d+b}{\ell}\right]$	Leverage	89.2	89.3
$\mathbb{E}\left[\frac{b}{b+d}\right]$	Share of time deposits	36.8	33.0
$\mathbb{E}[pr_\ell - ytm_{\mathrm{debt}}]$	Net interest margin	1.25	1.21
$\mathbb{E}[(1-p)\nu]$	Net charge-offs	0.19	0.20
Δ	Depreciation rate	2.5	2.5
Standard deviations			
$\sigma[(1-p)\nu]$	Net charge-offs	0.32	0.20
$\sigma\left[\frac{d+b}{\ell}\right]$	Leverage	3.6	1.3
$\sigma\left[\frac{b}{b+d}\right]$	Share of time deposits	16.8	8.1
$\sigma[ytm^b - ytm^d]$	Spread between time and demand deposits	0.21	0.18
$\sigma[pr_\ell - ytm_{\mathrm{debt}}]$	Net interest margin	0.24	0.10
Elasticity			
$\frac{\mathrm{cov}(400*R_\ell, \log(\ell))}{\mathrm{var}(400*R_\ell)}$	Loan Elasticity	-50	-36

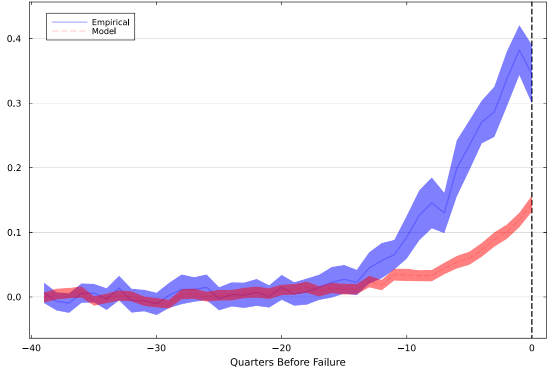
Model validation: profitability



- The model reproduces the decline in net interest margins before failure

Back to validation

Model fit: loan losses



- Deteriorating asset quality contributes to the gradual erosion of bank net worth.

Back to validation

Backup: resource constraint

$$\begin{aligned} div_t = {} & n_t + \tilde{q}_d(z, R'_\ell, d', b')d_{t+1} + \tilde{q}_b(z, R'_\ell, d', b')\left(b_{t+1} - (1-\lambda)b_t\right) \\ & - \ell^d(R_{\ell,t+1}, z_t, r^e_{t+1}) - Z(d_{t+1}, b_{t+1}, b_t) \\ & - \phi - (1-\gamma)\left(\frac{\max\{\ell_t - \ell^d(R_{\ell,t+1}, z_t, r^e_{t+1})}{\ell_t}\right)^2 \ell_t \\ & - \mathbf{1}_{\{b' < (1-\lambda)b\}}\left(q_b^{rf} - \tilde{q}_b(z, R'_\ell, d', b')\right). \end{aligned}$$

- Deposits fully mature every period.
- Only a fraction λ of time deposits matures each period.
- New issuance can dilute outstanding long-term creditors.

Back to bank problem

Backup: default thresholds

- $n^f(z, \ell, b)$: lowest cash-on-hand for continuation under normal funding.
- $n^c(z, \ell, b)$: lowest cash-on-hand for continuation during a rollover crisis.

$$n^f(z, \ell, b) \leq n^c(z, \ell, b).$$

- If $n < n^f$: fundamental default.
- If $n^f \leq n < n^c$: bank-run state.
- The sunspot selects default in the run region.

Back to environment

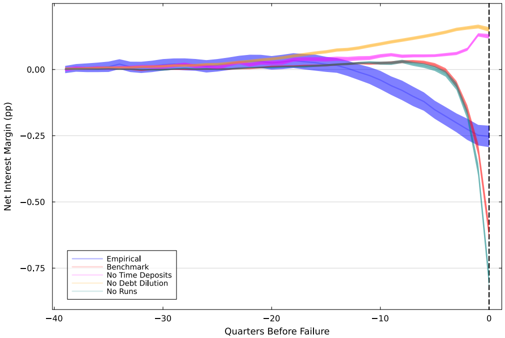
Backup: debt covenant counterfactual

$$CP(z, R'_\ell, d', b') = b(1 - \lambda)\left[q_b^{rf} - q_b(z, R'_\ell, d', b')\right].$$

- Banks compensate existing time-deposit holders for the decline in the value of outstanding claims.
- This eliminates debt dilution by construction.

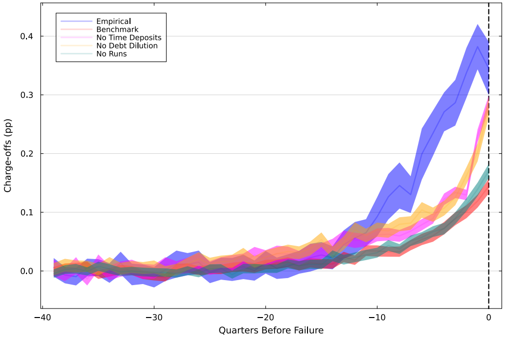
Back to counterfactual

Counterfactual: profitability



- As banks become more levered and rely more on time deposits, higher funding costs largely offset the increase in lending rates, compressing net interest margins.

Counterfactual: loan losses



- Eliminating debt dilution shifts failures from run-driven to fundamental defaults.
- As a result, banks experience larger loan losses by the time they default.